REVIEW

Which health benefits of the Mediterranean diet are actually supported?

REFERENCES	TOKENS	VERIFICATION	COMPILED
97 verified	≈19.8k	Crossref	27 Aug 2026

ABSTRACT

Mediterranean dietary programmes have their strongest clinical evidence in cardiovascular prevention. In higher-risk adults, three endpoint trials and modern syntheses indicate fewer major cardiovascular events, although absolute benefits are modest and Mediterranean programmes are not clearly superior to every supported low-fat programme. Randomized prevention, prospective cohort, and biomarker evidence also align for type 2 diabetes, while established disease shows small improvements in glycaemia and body mass. Higher adherence is consistently associated with longer survival, but no mortality trial establishes causality. Cancer and dementia evidence is dominated by cohorts; the largest dedicated cognitive trial was null. Small depression trials are favourable but heterogeneous, and fatty-liver effects resemble low-fat diets unless calorie restriction, activity, or polyphenol enrichment is added. Changes in inflammation, lipids, metabolites, and gut microbial functions cannot substitute for clinical outcomes. Overall, the pattern is a supported cardiometabolic programme whose effectiveness depends on adherence and implementation, not a collection of therapeutic foods or a universal preventive treatment.

01 Introduction

The Mediterranean diet is unusual in nutrition research because several major claims have been tested against clinical endpoints rather than only weight, cholesterol, or self-reported disease. Evidence includes randomized controlled trials of primary and secondary cardiovascular prevention, long prospective follow-up, and mechanistic trials using biomarkers and food provision. Yet the label covers variable interventions, and mortality, cancer, dementia, and mental-health evidence remains largely observational. The central question is not whether the pattern is "healthy" in the abstract, but which outcomes it changes, by how much, and under which comparison conditions.[1,2,3]

The distinction between a diet and a supported programme is consequential. Endpoint trials supplied foods, dietitian contact, group sessions, and monitoring, while newer studies add energy restriction, physical activity, or behavioural support.[4,5,6] Broad dietary claims, by contrast, draw far more often on cohort data and remain vulnerable to healthy-user confounding.[3] The throughline is therefore an evidence gradient: causal claims weaken as the review moves from supported cardiometabolic programmes and randomized events to observational associations and surrogate mechanisms.

02 Definition and measurement

The intervention is a family of related eating patterns rather than a prescribed menu. Reviews consistently identify a plant-rich core—vegetables, fruit, legumes, nuts, minimally refined cereals—combined with olive oil as the main added fat, regular fish, and lower consumption of red or processed meat and sweets. Dairy, eggs, alcohol, portion sizes, and frequencies vary across historical and modern definitions.[7,8] The Seven Countries Study helped formulate the hypothesis through long-term contrasts among populations, but its ecological and cohort observations were not randomized evidence.[9]

Most studies convert the pattern into an adherence score. Some scores award points relative to sex-specific or study-specific medians, while others use fixed intake targets; trials often use the 14-item Mediterranean Diet Adherence Screener (MEDAS).[10,11] These tools make a whole-diet exposure tractable, but they are not interchangeable: a two-point difference can represent different food substitutions in different datasets, and scoring systems developed in Mediterranean populations may classify non-Mediterranean diets differently.[12] Direct comparison within the MCC-Spain study found limited agreement among adherence indexes, showing that instrument choice can change who is labelled highly adherent.[13]

Synthesis quality compounds this problem. Across 24 cardiovascular reviews, complete satisfaction of adapted methodological criteria ranged from 8% to 75%; search details, participant characteristics, or appropriate pooling methods were often missing.[14] That label does not resolve variation in diet definition, comparator, support, or outcome measurement.

The stable core, variable definitions, and unsupported shortcuts are separated in Figure 1.

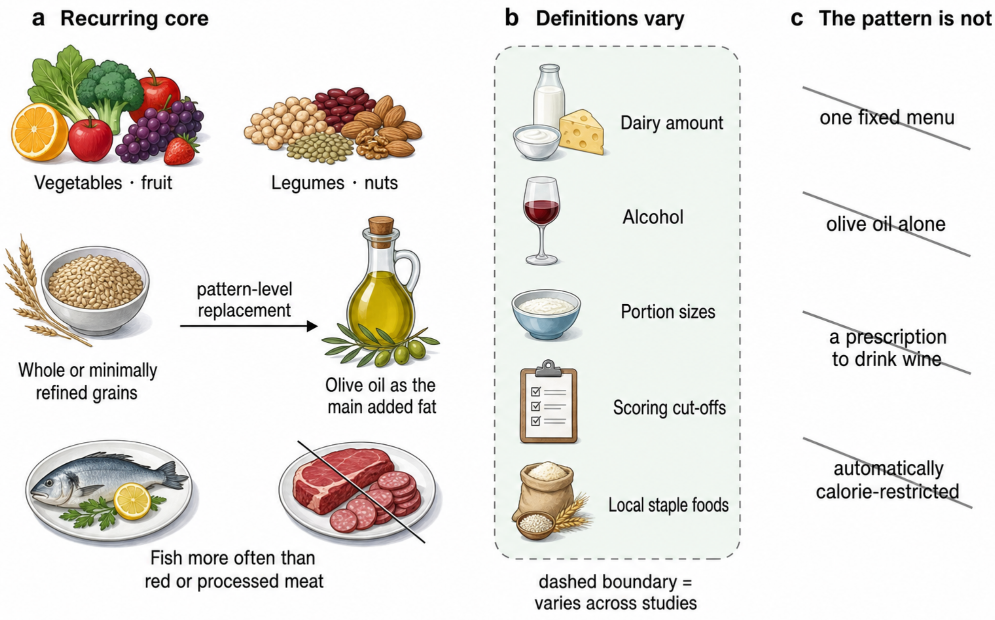

Figure 1. A Mediterranean diet is a pattern, not one fixed menu. The recurring food-group pattern is separated from definition-dependent choices and unsupported single-ingredient or fixed-menu interpretations. The replacement arrow is pattern-level synthesis, not a serving prescription or evidence that one component causes the outcomes.[7,8,12,15]

This instability sets the outcome standard: effects belong to the tested pattern and support package, not to the label or one food.

03 Cardiovascular endpoint trials

PREDIMED provides the largest randomized primary-prevention dataset, but only the corrected republication is usable evidence. A statistical investigation of 5,087 randomized trials detected improbable baseline distributions, prompting scrutiny of prominent reports.[16] PREDIMED investigators then identified household members enrolled without individual randomization and two sites with protocol departures. The 2013 report was retracted; the 2018 republication avoided assuming fully individual randomization and added exclusions and cluster-robust sensitivity analyses.[4]

In the corrected analysis, 7,447 adults at high cardiovascular risk were followed for a median of 4.8 years. Compared with low-fat advice, the primary composite of myocardial infarction, stroke, or cardiovascular death had a hazard ratio (HR) of 0.69 (95% confidence interval (CI) 0.53–0.91) for the extra-virgin-olive-oil arm and 0.72 (0.54–0.95) for the nut arm. Excluding 1,588 participants with known or suspected allocation departures yielded essentially the same estimates.[4] The correction preserves a clinically important result while reducing the certainty that would attach to an entirely conventional individual randomized controlled trial.

Two secondary-prevention trials extend the finding beyond one study. The Lyon Diet Heart Study randomized 605 people after myocardial infarction and reported cardiac death or nonfatal infarction in 14 Mediterranean-pattern participants and 44 controls over 46 months.[17] Its effect was strikingly large and has not been reproduced at the same magnitude. CORDIOPREV randomized 1,002 patients with established coronary disease to Mediterranean or low-fat diets; after seven years, 87 versus 111 participants experienced a major event, corresponding to adjusted hazard ratios around 0.72–0.75.[5] The three settings therefore agree on direction, while trial size, intervention composition, and effect magnitude differ.

TRIAL	SETTING	COMPARISON	MAIN ENDPOINT RESULT
PREDIMED republication	High-risk primary prevention; n=7,447	Olive oil or nuts vs low-fat advice	HR 0.69 and 0.72 for major cardiovascular events[4]
Lyon Diet Heart	Post-infarction; n=605	Mediterranean pattern vs prudent Western diet	Cardiac death/nonfatal infarction: 14 vs 44[17]
CORDIOPREV	Coronary disease; n=1,002	Mediterranean vs low-fat diet	87 vs 111 major events; adjusted HR about 0.72–0.75[5]

The exact trial values and the limits on comparing them are shown in Figure 2.

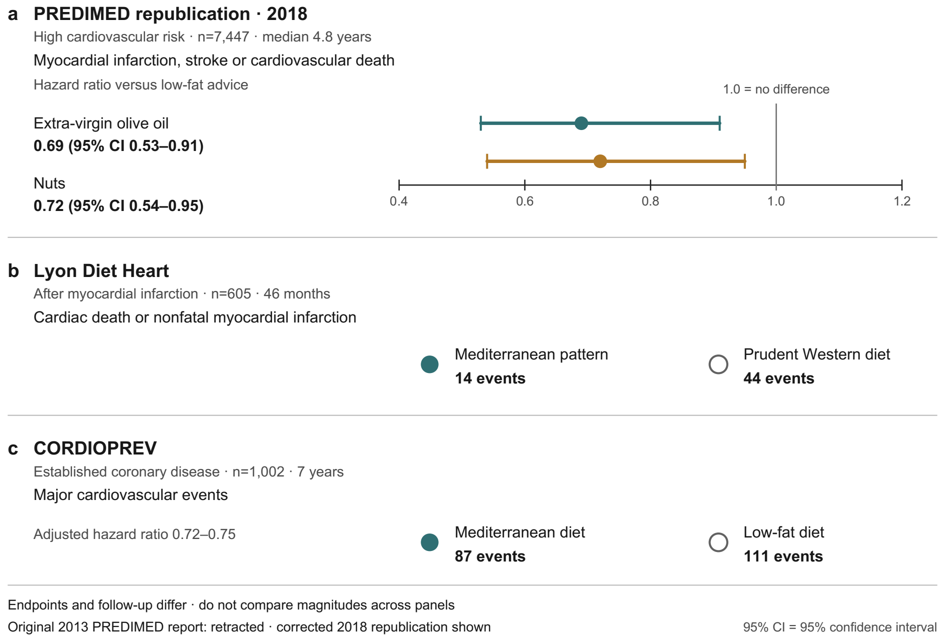

Figure 2. Three endpoint trials support cardiovascular benefit. The proportional panel shows corrected PREDIMED hazard ratios and confidence intervals, while equal-size markers accompany the reported Lyon Diet Heart and CORDIOPREV event counts. Populations, comparators, endpoints, and follow-up differ, so cross-panel magnitude comparisons are invalid; the original 2013 PREDIMED report is retracted, and only the corrected 2018 republication is plotted.[4,17,5,1]

These trials are the review's causal anchor, but their differences support a programme-level direction rather than one universal effect size.

04 Comparative cardiovascular evidence

The most informative modern synthesis compares supported programmes with both minimal advice and other active diets. A network meta-analysis of 40 trials and 35,548 higher-risk participants estimated that Mediterranean programmes reduced all-cause mortality (odds ratio 0.72), cardiovascular mortality (0.55), stroke (0.65), and nonfatal infarction (0.48) versus minimal intervention. At intermediate baseline risk, the estimates corresponded to about 17 fewer deaths, 13 fewer cardiovascular deaths, seven fewer strokes, and 17 fewer infarctions per 1,000 people over five years.[1] However, low-fat programmes also reduced mortality and infarction, and there was no convincing advantage of Mediterranean over low-fat programmes for those endpoints. The result supports structured dietary care more strongly than dietary exceptionalism.

Earlier syntheses were more cautious. Six randomized trials with 10,950 participants produced a risk ratio of 0.63 (95% CI 0.53–0.75) for major vascular events but a null 1.00 (0.86–1.15) for all-cause mortality; excluding one trial with serious integrity concerns erased several favourable secondary findings.[18] A Cochrane review of 30 trials and 12,461 participants emphasized low-to-moderate certainty, heterogeneous definitions, sparse events, and weak adverse-event reporting.[19] Another review found benefits from unrestricted-fat Mediterranean patterns but noted that clinical-event evidence rested on few trials.[20]

Newer reviews still find fewer cardiovascular events but remain structurally dependent on PREDIMED.[21] The 2026 Italian-guideline syntheses covered 87 primary-prevention studies with more than 1.4 million participants and 19 secondary-prevention studies with more than 91,000; they reported lower incident and recurrent risk but combined randomized and cohort estimates.[22,23] Risk-factor differences from low-fat diets are smaller than event contrasts, and active-diet differences are narrower than comparisons with usual care.[24]

The cardiovascular case is therefore strongest against minimal support, less distinctive against another supported diet, and less direct once the evidence shifts to cohort longevity.

05 Mortality and longevity

Prospective cohorts show a dose-response association with survival. Each two-point score increase corresponded to 25% lower mortality in 22,043 Greek adults and 8% in 74,607 older Europeans.[25,26] Geographic replication thus preserved the association at a smaller magnitude.

Pooling 12 cohorts and 1.57 million participants gave an all-cause mortality risk ratio of 0.91 per two score points; later syntheses remained near a 9%–10% reduction despite varying scores and follow-up.[27,28,29] A 2026 meta-analysis of 54 cohorts, 1.83 million participants, and 346,034 deaths found risk ratio 0.96 (95% CI 0.95–0.97) per point, with moderate certainty.[2] This graded association does not show that one menu reduces mortality by a fixed percentage.

US cohorts support transfer: top healthy-pattern categories were associated with total mortality HR 0.82 and cardiovascular disease HR 0.83.[30,31] Among 25,315 women followed for up to 25 years, top adherence corresponded to mortality HR 0.77, only partly explained by measured metabolic and inflammatory pathways.[32] In the Moli-sani diabetes cohort, mortality HR was 0.63 per two score points.[33]

Objective measures reduce reporting concerns but remain observational. In 642 older Italians followed for 20 years, a biomarker score predicted all-cause mortality HR 0.72 and cardiovascular mortality HR 0.55 for the highest versus lowest tertile, outperforming questionnaire scoring. Agreement was low, biomarkers were measured only at baseline, and residual confounding remained possible.[34] Mortality evidence is therefore consistent and dose-responsive, but its causal magnitude remains uncertain without a trial powered for that endpoint.

06 Diabetes, weight, and metabolic health

Type 2 diabetes is the non-cardiovascular outcome where experimental, prospective, and biomarker evidence align most closely. Both PREDIMED interventions lowered incidence versus low-fat advice in participants free of diabetes.[35] Up to eight annual adherence measurements yielded HR 0.80 (95% CI 0.70–0.92) per two MEDAS points and 0.46 (0.25–0.83) for highest versus lowest adherence.[36] Because attained adherence is partly behavioural, these estimates show a gradient within a trial context.

PREDIMED-Plus tested an intensive weight-loss package. Among 4,746 adults free of diabetes, a 30%-energy-restricted Mediterranean diet plus activity reduced six-year incidence by 31% relative to unrestricted Mediterranean diet: 9.5% versus 12.0%.[6] The comparison cannot isolate diet, energy deficit, exercise, and behavioural contact. In a 1,521-person imaging substudy, the package reduced total and visceral fat and attenuated relative lean-mass decline over three years, with partial weight regain after year one.[37]

A score based on five carotenoids and 24 fatty acids identified the Mediterranean arm of a feeding trial with C-statistic 0.88, then predicted diabetes HR 0.71 (95% CI 0.65–0.77) per standard deviation in 22,202 EPIC-InterAct participants, versus 0.90 for self-report.[38] It may capture other foods and residual confounding, but weakens reporting bias as the sole explanation. A Mediterranean-lifestyle index predicted HR 0.70 for highest versus lowest quartile in 112,493 UK Biobank adults, although it included activity, rest, and social habits.[39]

Prospective-study reviews estimate 19%–23% lower diabetes incidence with higher adherence, an observational result consistent with but less causal than the trials.[40,41] Treatment effects in established diabetes are smaller; reviews find modest glycaemic improvement versus comparator diets. Across 11 trials, glycated haemoglobin (HbA1c) was 0.31 percentage points lower, fasting glucose 0.85 mmol/L lower, and body mass index 0.83 kg/m² lower.[42] These are adjunctive changes, not evidence that diet replaces medication.

Weight-loss comparisons also argue against uniqueness. DIRECT found two-year losses of 4.4 kg with Mediterranean, 2.9 kg with low-fat, and 4.7 kg with low-carbohydrate diets in 322 adults.[43] Across 121 trials, modest six-month losses diminished by 12 months; Mediterranean programmes retained some cardiovascular risk-factor improvement but no superior long-term weight loss.[44] A 50-study synthesis estimated changes of about −2.4 mm Hg systolic pressure, −6 mg/dL triglycerides, −4 mg/dL glucose, and under 1 cm waist circumference; across 57 trials and 36,983 participants, direction favoured Mediterranean diet for 18 of 28 metabolic measures, but heterogeneity was often high.[45,46] A green variant reduced visceral fat by 14.1%, versus 6.0% with standard Mediterranean diet and 4.2% with healthy guidance, but added walnuts, green tea, Mankai, energy restriction, and activity in a workforce that was 88% male.[47]

Adding olive oil does not reproduce the whole pattern: a 2024 meta-analysis of enriched Mediterranean trials found no consistent cardiometabolic advantage except a very small triglyceride reduction.[15] In pre-existing metabolic disease, a 69-study synthesis found modest improvements in body mass index, waist, low-density lipoprotein (LDL) cholesterol, triglycerides, fasting glucose, insulin resistance, and C-reactive protein (CRP), but inconsistent HbA1c and lean-mass results.[48]

Diabetes repeats the cardiovascular pattern: the strongest benefits belong to the supported diet, while weight and risk-factor changes are shared with other effective diets.

07 Cancer

Cancer is where breadth of association most clearly exceeds causal depth. An umbrella review judged evidence comparatively robust for overall cancer incidence and mortality but only suggestive or weak for many individual sites.[49] The largest updated synthesis included 117 studies and 3.20 million participants. High adherence was associated with cancer mortality risk ratio 0.87, colorectal cancer 0.83, breast cancer 0.94, and larger reductions for some less common sites, yet most site-specific comparisons were rated low or very low certainty.[50]

Repeated meta-analyses find the same broad direction but share the same observational inputs, scoring heterogeneity, and residual lifestyle confounding.[51] Dose-response analysis supports lower colorectal risk, and a cohort/meta-analysis of postmenopausal breast cancer found the inverse association concentrated in estrogen-receptor-negative tumours.[52,53] These subgroup patterns can be biologically interesting without being intervention effects.

Randomized evidence consists mainly of one small secondary outcome. In PREDIMED, the olive-oil group experienced fewer invasive breast cancers than low-fat controls, HR 0.32 (95% CI 0.13–0.79), but only 35 cases occurred and cancer was not the trial's primary endpoint.[54] A breast-cancer umbrella review reaches a favourable overall interpretation while acknowledging the predominance of observational evidence.[55] Olive-oil intake itself is associated with lower cardiovascular disease, diabetes, and mortality, but its pooled cancer estimate is null at risk ratio 0.94 (0.86–1.03).[56] The cancer literature therefore supports prevention hypotheses and a generally plant-rich pattern, not a claim that Mediterranean diet treatment prevents specific cancers.

08 Cognition and dementia

Prospective studies associate Mediterranean adherence with lower cognitive decline. A 2025 meta-analysis of 23 studies reported HR 0.82 for cognitive impairment, 0.89 for dementia, and 0.70 for Alzheimer disease, while documenting heterogeneity, possible publication bias, and limited non-Western representation.[57] A separate updated synthesis also found inverse dementia and Alzheimer associations.[58] UK Biobank provides a useful absolute-risk estimate: among 60,298 adults, high adherence corresponded to dementia HR 0.77 and absolute risk 1.18% versus 1.73% over 9.1 years, without measurable interaction with polygenic risk.[59]

Earlier reviews combined favourable cohorts with small trials. One synthesis found mild cognitive impairment risk ratio 0.75 and Alzheimer disease 0.71 in cohorts; two trials yielded an episodic-memory standardized mean difference of 0.20.[60] Another meta-analysis reached a similar observational conclusion.[61] A PREDIMED-Navarra substudy reported better global cognition after 6.5 years, but it was small and inherited the allocation limitations of the parent trial.[62]

The largest dedicated trial sharply lowers expectations. The three-year MIND trial randomized 604 older adults with suboptimal diets and obesity history; both intervention and control were calorie-restricted. Global cognition differed by only 0.035 standard deviations (95% CI −0.022 to 0.092), and brain magnetic resonance imaging (MRI) outcomes did not differ.[63] A MIND-diet meta-analysis remains favourable because its pooled estimate is dominated by cohorts, whereas a 2024 PREVENT analysis found no association with cognition or neuroimaging.[64,65] The most defensible synthesis is that long-term adherence may mark lower dementia risk, but a three-year supported MIND intervention did not improve cognition in relatively healthy older volunteers.

09 Depression and mental health

Prospective cohorts associate Mediterranean adherence with lower incident depression. Meta-analyses report an inverse association of roughly one-third for the highest adherence groups, although reverse causation is difficult to exclude because early depression can alter shopping, appetite, activity, and diet quality.[66,67] The observational result is therefore compatible with benefit without proving treatment efficacy.

Small randomized trials provide a causal signal but an unstable magnitude. SMILES randomized 67 adults with major depression to dietary support or befriending-style social support for 12 weeks. The diet group improved 7.1 points more on the Montgomery-Åsberg Depression Rating Scale, and remission was 32% versus 8%.[68] HELFIMED enrolled 152 adults and found greater symptom improvement with group Mediterranean education plus fish-oil supplementation than with social groups, a design that cannot isolate the diet.[69] AMMEND extended the result to 72 young men over 12 weeks, again in a small, selected population.[70]

Pooling magnifies the uncertainty. Five trials with 1,507 participants yielded symptom standardized mean difference −0.53, but heterogeneity was 87%, the prediction interval crossed zero, and certainty was low.[71] A 2025 review identified 104 mental-health studies—88 observational and only 16 interventional—and found favourable associations with depression, anxiety, stress, quality of life, and well-being.[72] Direction is consistent enough to justify larger trials, but current evidence does not identify a stable antidepressant effect or support replacing established care.

10 Fatty liver

Mediterranean interventions reduce hepatic fat, but direct comparison weakens claims of unique superiority. In a 12-week randomized trial, ad libitum Mediterranean and low-fat diets both reduced steatosis without a decisive between-diet difference.[73] In DIRECT, decreased hepatic fat statistically mediated part of the Mediterranean diet's cardiometabolic advantage over low fat, but mediation cannot identify the responsible foods.[74]

The largest imaging effects come from an enriched, energy-restricted variant. DIRECT-PLUS randomized 294 adults to healthy guidance, standard Mediterranean, or green Mediterranean diets, with physical activity in every arm. Relative liver-fat reduction was 12.2%, 19.6%, and 38.9%, respectively, after 18 months; the green arm added walnuts, green tea, Mankai, and further red-meat restriction.[75] This shows that a complex enriched programme can change liver fat, not that olive oil alone treats liver disease.

Meta-analyses explain the mixed interpretation. Earlier randomized-trial pooling found improvement in steatosis and insulin resistance and small, inconsistent enzyme changes.[76,77] A 2024 head-to-head review concluded that Mediterranean and low-fat diets had similar short-term effects on liver enzymes and fat content.[78] An Asian-adapted randomized trial suggests the approach can be transferred culturally, but adaptation changes the tested foods and long-term clinical liver outcomes remain unavailable.[79]

Fatty liver repeats the weight pattern: intermediate outcomes improve, but active-diet comparisons and bundled interventions limit claims of Mediterranean-specific efficacy.

11 Biological pathways

Mechanistic evidence supports plausibility through several pathways rather than one "active ingredient." Trial syntheses report small changes in endothelial function and selected inflammatory markers, with high heterogeneity across assays and populations.[80,81] A 2026 review of 33 randomized trials and 3,476 participants found reductions in high-sensitivity CRP, interleukin-6, and interleukin-17, but not conventional CRP, interleukin-10, tumour-necrosis factor, or total antioxidant capacity.[82] Reviews therefore implicate fat quality, plant polyphenols, fibre fermentation, inflammation, endothelial function, and insulin sensitivity without establishing one required component.[83]

Controlled feeding and isocaloric designs separate dietary composition from weight loss. In an eight-week trial of 82 overweight or obese adults, a weight-maintaining Mediterranean diet lowered total cholesterol and altered faecal bile acids, microbial functions, and metabolites. Glucose, insulin, and CRP did not change.[84] In older adults, the one-year NU-AGE intervention shifted microbial taxa and functions that covaried with lower frailty and inflammation; the analysed microbiome subset included 612 participants across five countries.[85] A separate obesity trial found that both Mediterranean and low-fat healthy diets modified gut communities alongside improved insulin sensitivity.[86]

Molecular profiles are also becoming exposure measures. Plasma metabolomic patterns tracked Mediterranean adherence and subsequent cardiovascular risk, though they cannot by themselves prove mediation.[87] In DIRECT-PLUS, baseline bile acids and particular microbial species modified lipid and adiposity responses, while a PREDIMED-Plus analysis connected intervention-related microbiome and metabolome shifts with risk-factor changes.[88,89] These studies clarify how people may respond differently; they do not upgrade surrogate changes into evidence for fewer cancers, dementia cases, or deaths.

12 Implementation, cost, and sustainability

Transfer requires adapting the plant-forward structure and fat quality to local staples, availability, cooking, and culture.[90,91] Scores built around olive oil, wine, and regional medians may misclassify an equally plant-rich pattern elsewhere. Implementation concerns what replaces refined grains, processed meat, and saturated fat, not geographic authenticity.

The intervention package, adherence conditions, and the causal stopping point are traced in Figure 3.

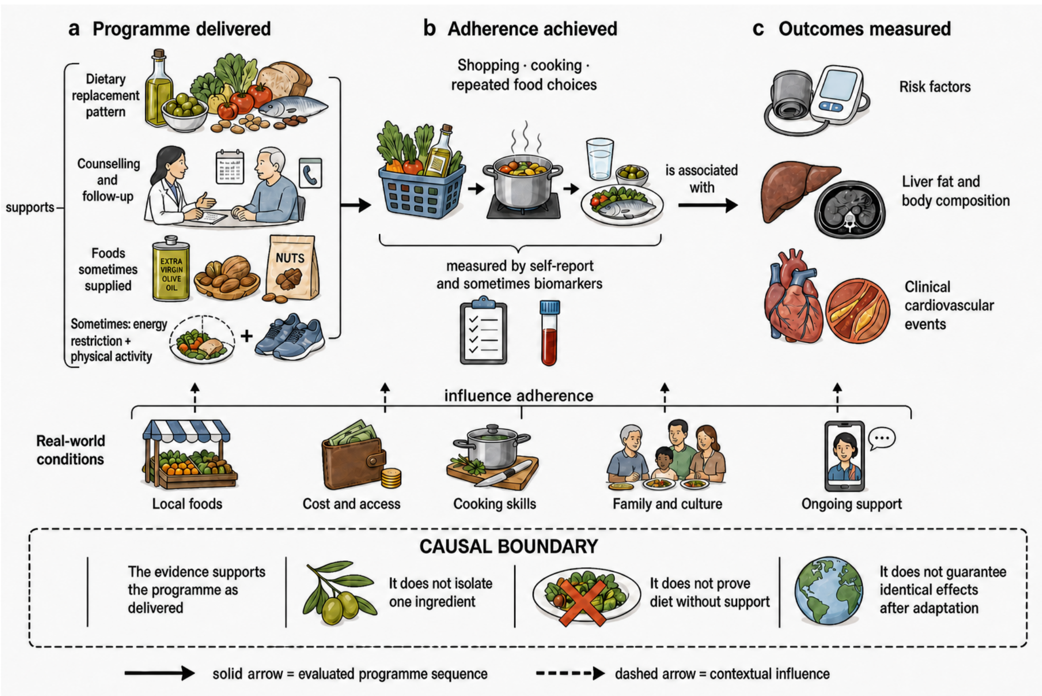

Figure 3. Benefits belong to the supported programme as delivered. Trials evaluate a replacement pattern with varying counselling, food provision, and lifestyle support; real-world conditions influence adherence before risk factors, imaging outcomes, or clinical events are measured. The arrows express programme logic rather than a quantified mediation model, and the dashed boundary prevents attributing the package to one ingredient, unsupported diet alone, or identical effects after cultural adaptation.[4,92,93,94,6]

Implementation is part of efficacy: trial benefits belong to supported replacement patterns and may change with support, affordability, or cultural fit.

Supported efficacy exceeds unsupported adoption. MedEx-UK improved adherence and acceptability in older adults at dementia risk but included behavioural maintenance and physical activity.[92] Cost, access, cooking skills, family norms, and knowledge remain barriers.[93] PREDIMED supplied oil or nuts; PREDIMED-Plus added dietitian contact and activity coaching.[4,6] Benefits may decline when those supports disappear.

Economic and environmental effects depend on assumptions. A secondary-prevention model found Mediterranean diet plus activity cost-effective but inherited adherence and event estimates from other data.[95] Of 35 sustainability studies, mostly models, 91% favoured the pattern environmentally; gains centred on less ruminant meat, while water demand and local production changed rankings.[94] Med Diet 4.0 likewise treats health, environment, culture, and local economy as linked rather than interchangeable.[96]

Alcohol requires a modern qualification. Traditional scores often award a point for moderate wine, but alcohol raises cancer, dependence, and injury risk. A contemporary review argues that wine is not necessary to preserve the pattern's benefits and should not be recommended to non-drinkers.[97] The therapeutic interpretation is therefore vegetables, legumes, whole grains, nuts, fish, and unsaturated fats—not an instruction to begin drinking.

13 Overall certainty

OUTCOME	CURRENT CONCLUSION	MAIN REASON CONFIDENCE IS LIMITED
Major cardiovascular events	Probably reduced in higher-risk adults, particularly versus minimal support	Few endpoint trials; corrected PREDIMED carries much of primary-prevention weight[1]
Type 2 diabetes	Probably prevented; established disease improves modestly	Intensive programmes combine diet, energy restriction, activity, and support[6]
All-cause mortality	Consistent 4% lower risk per adherence point in cohorts	No randomized mortality confirmation; residual healthy-user bias[2]
Weight and risk factors	Small favourable changes	Similar benefits occur with other supported healthy diets[44]
Cancer	Several inverse site-specific associations	Predominantly observational; one small randomized secondary signal[50]
Cognition	Cohorts favourable, largest dedicated trial null	Active control, limited duration, exposure confounding[63]
Depression	Small trials favourable	Low certainty, high heterogeneity, weak blinding[71]
Fatty liver	Liver fat decreases	Short-term equivalence to low-fat diets; few clinical outcomes[78]

Directness declines from randomized cardiovascular and diabetes evidence through cohort mortality, predominantly observational cancer and dementia evidence, limited depression trials, and surrogate liver outcomes. Mechanisms make the cardiometabolic findings coherent but cannot close those causal gaps.

This uneven directness and certainty are mapped in Figure 4.

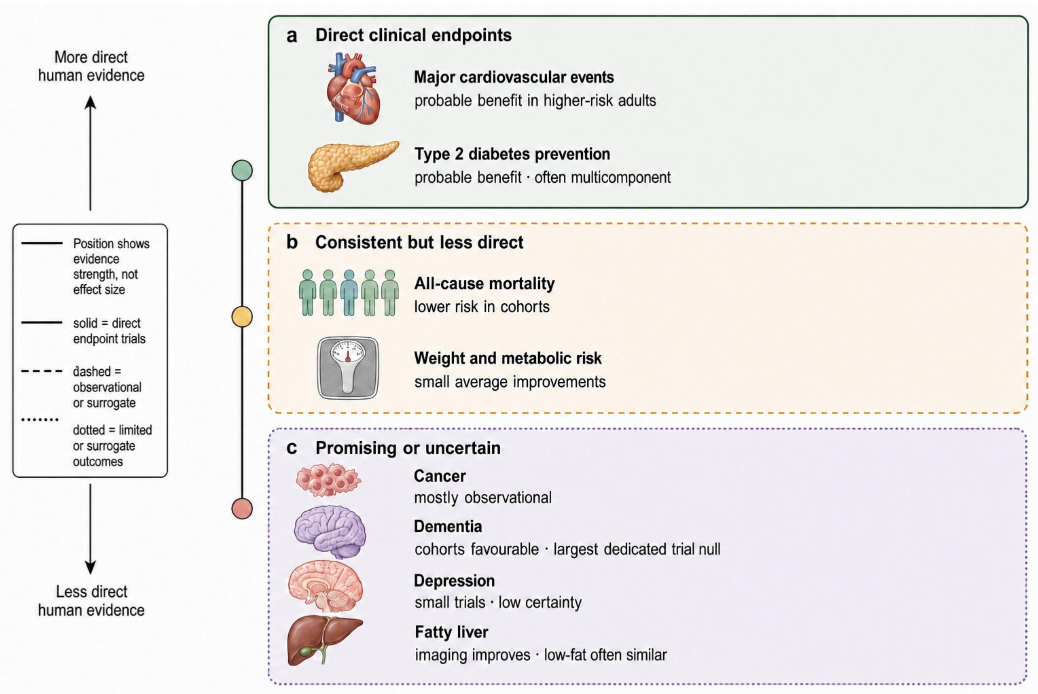

Figure 4. The evidence is strongest for cardiometabolic outcomes. Outcomes are arranged by human evidence directness and certainty, from clinical endpoint trials through consistent but less-direct evidence to limited or surrogate outcomes. This is a narrative-review synthesis rather than a formal GRADE score: position, marker area, and colour do not encode effect size or an individual's expected benefit.[1,6,2,50,63]

Across outcomes, greater distance from randomized clinical endpoints requires greater qualification: plausibility extends beyond demonstrated clinical benefit.

14 Conclusion

The cross-cutting pattern is an evidence gradient, not one verdict on a diet. Supported Mediterranean programmes probably reduce major cardiovascular events in higher-risk adults and prevent type 2 diabetes, especially compared with minimal advice. Mortality, cancer, and dementia conclusions rely increasingly on cohorts, while mechanisms and imaging support plausibility without proving disease prevention. Supported low-fat diets also narrow head-to-head differences, and several favourable interventions bundle diet with energy restriction, activity, food provision, or counselling.

The defensible recommendation is therefore a culturally workable, plant-forward replacement pattern with credible cardiometabolic value, not isolated foods or nutrients treated as medicines. Depression remains a heterogeneous small-trial signal, and liver-fat improvement has not established Mediterranean-specific clinical benefit. Long-term randomized trials with active comparators, clinical endpoints, and intervention arms that separate diet composition from energy restriction, activity, and behavioural contact would show how much of the wider promise is causal and how much belongs to adherence support.

References

97 · VERIFIED VIA CROSSREF

1. **Karam G, Agarwal A, Sadeghirad B, Jalink M, Hitchcock CL, Ge L, Kiflen R, Ahmed W, Zea AM, Milenkovic J, Chedrawe MA, Rabassa M, El Dib R, Goldenberg JZ, Guyatt GH, Boyce E, Johnston BC (2023)** Comparison of seven popular structured dietary programmes and risk of mortality and major cardiovascular events in patients at increased cardiovascular risk: systematic review and network meta-analysis. *BMJ*. https://doi.org/10.1136/bmj-2022-072003
2. **Nucci D, Ragusa FS, Mazza E, Veronese N, Gianfredi V, Volpe M, Maggi S, Onder G, Sieber C, Zanetti M, Silano M, Chernykh A, Lorenzo FD, Romiti GF, Schirò P, Ungar A, Maggi S, Volpe M, Zanetti M, Onder G, Veronese N, Gianfredi V, Nucci D, Silano M, Rossi L, Antonella B, Paolo C, Maria CA, Matteo CM, Giovambattista D, Ferruccio G, Emilia G, Antonella L, Francesco L, Francesco M, Ida MM, Gerardo M, Barbara P, Domenico R, Elena S, Massimiliano S, Immacolata SM, Ersilia T, Andrea U, Roberto V, Giovanni Z, Elena A, Fiorella A, Letizia B, Federico B, Alberto B, Ilarj BM, Carola B, Stefania C, Ilaria C, Massimiliano C, Irene C, Amanda C, Elisa C, Anastasia C, Martina C, Martina C, Giuseppe CAF, Laura C, Marco C, Stefano C, Lucilla C, Chiara D, Andrea D, Francesco DL, Lorenzo E, Maurizio F, Enrica F, Martina F, Armando F, Lucia G, Cristina G, Alessandro G, Paola G, Federica L, Fabrizia L, Elisa M, Alessandro M, Pina MM, Alessia M, Alisha M, Giovanna M, Barbara P, Stefania P, Damiano P, Cristina PM, Costanza P, Saverio RF, Cristiana R, Francesco RG, Alessia S, Piero S, Sara S, Dario TG, Ilaria T, Filippo V, Fernanda V, Marilena V, Salvatore Z, Morsella A, Limongi F, Ragusa FS, Fontana L, Laviano A, Sieber C (2026)** Adherence to the Mediterranean diet and all-cause mortality: A systematic review and meta-analysis within the Italian National Guidelines “La Dieta Mediterranea”. *Nutrition*. https://doi.org/10.1016/j.nut.2026.113189
3. **English LK, Ard JD, Bailey RL, Bates M, Bazzano LA, Boushey CJ, Brown C, Butera G, Callahan EH, de Jesus J, Mattes RD, Mayer-Davis EJ, Novotny R, Obbagy JE, Rahavi EB, Sabate J, Snetselaar LG, Stoody EE, Van Horn LV, Venkatramanan S, Heymsfield SB (2021)** Evaluation of Dietary Patterns and All-Cause Mortality. *JAMA Network Open*. https://doi.org/10.1001/jamanetworkopen.2021.22277
4. **Estruch R, Ros E, Salas-Salvadó J, Covas MI, Corella D, Arós F, Gómez-Gracia E, Ruiz-Gutiérrez V, Fiol M, Lapetra J, Lamuela-Raventos RM, Serra-Majem L, Pintó X, Basora J, Muñoz MA, Sorlí JV, Martínez JA, Fitó M, Gea A, Hernán MA, Martínez-González MA (2018)** Primary Prevention of Cardiovascular Disease with a Mediterranean Diet Supplemented with Extra-Virgin Olive Oil or Nuts. *New England Journal of Medicine*. https://doi.org/10.1056/nejmoa1800389
5. **Delgado-Lista J, Alcala-Diaz JF, Torres-Peña JD, Quintana-Navarro GM, Fuentes F, Garcia-Rios A, Ortiz-Morales AM, Gonzalez-Requero AI, Perez-Caballero AI, Yubero-Serrano EM, Rangel-Zuñiga OA, Camargo A, Rodriguez-Cantalejo F, Lopez-Segura F, Badimon L, Ordovas JM, Perez-Jimenez F, Perez-Martinez P, Lopez-Miranda J, Alcala-Diaz JF, Almaden Peña Y, Aranda E, Arenas de Larriva AP, Badimon L, Badimon JJ, Blanco-Molina A, Blanco-Rojo R, Bolivar-Muñoz J, Caballero-Villarraso J, Camargo A, Chica J, Corina A, Criado-Garcia J, Cruz-Teno C, Daponte-Codina A, de Teresa Galvan E, Delgado-Casado N, Delgado-Lista J, Estruch R, Fernandez JM, Fernandez-Gandara C, Fuentes-Jimenez F, Garcia-Carpintero Fernandez-Pacheco S, Garcia-Rios A, Gomez-Delgado F, Gomez-Garduño A, Gomez-Luna P, Gomez-Luna MJ, Gonzalez-Guardia L, Gonzalez-Requero AI, Gutierrez-Mariscal FM, Haro-Mariscal CM, Jimenez-Lucena R, Jimenez-Morales AI, Leon-Acuña A, Lopez-Miranda J, Lopez-Segura F, Marin-Hinojosa C, Meneses Alvarez ME, Mesa-Luna D, Moya-Garrido MN, Muñoz-Carvajal I, Navarro-Martos V, Ochoa JJ, Ordovas JM, Ortiz-Minuesa JA, Ortiz-Morales AM, Pan M, Peña-Orihuela P, Perez-Caballero AI, Perez-Corral I, Perez-Jimenez F, Perez-Martinez P, Pi-Sunyer FX, Quintana-Navarro GM, Ramirez-Lara I, Rangel-Zuñiga OA, Rodriguez-Artalejo F, Rodriguez-Cantalejo F, Romero MA, Roncero-Ramos I, Ruano-Ruiz JA, Ruiz de Castroviejo J, Sanchez-Villegas P, Suarez de Lezo J, Suarez de Lezo J, Torres-Peña JD, Vals-Delgado C, Valverde R, Visioli F, Yubero-Serrano EM (2022)** Long-term secondary prevention of cardiovascular disease with a Mediterranean diet and a low-fat diet (CORDIOPREV): a randomised controlled trial. *The Lancet*. https://doi.org/10.1016/s0140-6736%2822%2900122-2
6. **Ruiz-Canela M, Corella D, Martínez-González MÁ, Babio N, Martínez JA, Forga L, Alonso-Gómez ÁM, Wärnberg J, Vioque J, Romaguera D, López-Miranda J, Estruch R, Santos-Lozano JM, Serra-Majem L, Bueno-Cavanillas A, Tur JA, Martín-Sánchez V, Riera-Mestre A, Delgado-Rodríguez M, Matía-Martín P, Vidal J, Vázquez C, Daimiel L, Buil-Cosiales P, Shyam S, Sorlí JV, Castañer O, García-Rios A, Torres-Collado L, Gómez-Gracia E, Zulet MÁ, Konieczna J, Casas R, Cano-Ibáñez N, Tojal-Sierra L, Bernal-López RM, Toledo E, García-Gavilán J, Fernández-Carrión R, Goday A, Arenas-Larriva AP, González-Palacios S, Schröder H, Ros E, Fitó M, Hu FB, Tinahones FJ, Salas-Salvadó J (2025)** Comparison of an Energy-Reduced Mediterranean Diet and Physical Activity Versus an Ad Libitum Mediterranean Diet in the Prevention of Type 2 Diabetes. *Annals of Internal Medicine*. https://doi.org/10.7326/annals-25-00388
7. **Davis C, Bryan J, Hodgson J, Murphy K (2015)** Definition of the Mediterranean Diet; A Literature Review. *Nutrients*. https://doi.org/10.3390/nu7115459
8. **Bach-Faig A, Berry EM, Lairon D, Reguant J, Trichopoulou A, Dernini S, Medina FX, Battino M, Belahsen R, Miranda G, Serra-Majem L (2011)** Mediterranean diet pyramid today. Science and cultural updates. *Public Health Nutrition*. https://doi.org/10.1017/s1368980011002515
9. **Menotti A, Puddu PE (2015)** How the Seven Countries Study contributed to the definition and development of the Mediterranean diet concept: A 50-year journey. *Nutrition, Metabolism and Cardiovascular Diseases*. https://doi.org/10.1016/j.numecd.2014.12.001
10. **Bach A, Serra-Majem L, Carrasco JL, Roman B, Ngo J, Bertomeu I, Obrador B (2006)** The use of indexes evaluating the adherence to the Mediterranean diet in epidemiological studies: a review. *Public Health Nutrition*. https://doi.org/10.1079/phn2005936
11. **Schröder H, Fitó M, Estruch R, Martínez-González MA, Corella D, Salas-Salvadó J, Lamuela-Raventós R, Ros E, Salaverría I, Fiol M, Lapetra J, Vinyoles E, Gómez-Gracia E, Lahoz C, Serra-Majem L, Pintó X, Ruiz-Gutierrez V, Covas MI (2011)** A Short Screener Is Valid for Assessing Mediterranean Diet Adherence among Older Spanish Men and Women. *The Journal of Nutrition*. https://doi.org/10.3945/jn.110.135566
12. **Hutchins-Wiese HL, Bales CW, Porter Starr KN (2022)** Mediterranean diet scoring systems: understanding the evolution and applications for Mediterranean and non-Mediterranean countries. *British Journal of Nutrition*. https://doi.org/10.1017/s0007114521002476
13. **Olmedo-Requena R, González-Donquiles C, Dávila-Batista V, Romaguera D, Castelló A, Molina de la Torre AJ, Amiano P, Dierssen-Sotos T, Guevara M, Fernández-Tardón G, Lozano-Lorca M, Alguacil J, Peiró R, Huerta JM, García-Lavedan E, Aragonés N, Fernández-Villa T, Solans M, Gómez-Acebo I, Castaño-Vinyals G, Kogevinas M, Pollán M, Martín V (2019)** Agreement among Mediterranean Diet Pattern Adherence Indexes: MCC-Spain Study. *Nutrients*. https://doi.org/10.3390/nu11030488
14. **Huedo-Medina TB, Garcia M, Bihuniak JD, Kenny A, Kerstetter J (2016)** Methodologic quality of meta-analyses and systematic reviews on the Mediterranean diet and cardiovascular disease outcomes: a review. *The American Journal of Clinical Nutrition*. https://doi.org/10.3945/ajcn.115.112771
15. **Keshani M, Sadeghi N, Tehrani SD, Ahmadi AR, Sharma M (2024)** Mediterranean diet enriched with olive oil shows no consistent benefits on cardiometabolic and anthropometric parameters: a systematic review with meta-analysis of randomized controlled trials. *European Journal of Nutrition*. https://doi.org/10.1007/s00394-024-03478-9
16. **Carlisle JB (2017)** Data fabrication and other reasons for non-random sampling in 5087 randomised, controlled trials in anaesthetic and general medical journals. *Anaesthesia*. https://doi.org/10.1111/anae.13938
17. **de Lorgeril M, Salen P, Martin JL, Monjaud I, Delaye J, Mamelle N (1999)** Mediterranean Diet, Traditional Risk Factors, and the Rate of Cardiovascular Complications After Myocardial Infarction. *Circulation*. https://doi.org/10.1161/01.cir.99.6.779
18. **Liyanage T, Ninomiya T, Wang A, Neal B, Jun M, Wong MG, Jardine M, Hillis GS, Perkovic V (2016)** Effects of the Mediterranean Diet on Cardiovascular Outcomes—A Systematic Review and Meta-Analysis. *PLOS ONE*. https://doi.org/10.1371/journal.pone.0159252
19. **Rees K, Takeda A, Martin N, Ellis L, Wijesekara D, Vepa A, Das A, Hartley L, Stranges S (2019)** Mediterranean-style diet for the primary and secondary prevention of cardiovascular disease. *Cochrane Database of Systematic Reviews*. https://doi.org/10.1002/14651858.cd009825.pub3
20. **Bloomfield HE, Koeller E, Greer N, MacDonald R, Kane R, Wilt TJ (2016)** Effects on Health Outcomes of a Mediterranean Diet With No Restriction on Fat Intake. *Annals of Internal Medicine*. https://doi.org/10.7326/m16-0361
21. **Sebastian SA, Padda I, Johal G (2024)** Long-term impact of mediterranean diet on cardiovascular disease prevention: A systematic review and meta-analysis of randomized controlled trials. *Current Problems in Cardiology*. https://doi.org/10.1016/j.cpcardiol.2024.102509

22. **Volpe R, Ciccone MM, Pala B, Barbarano F, Camastra S, Caprio M, Casirati A, Ferrera A, Galletti F, Greatti A, Mollica MP, Paolillo S, Parretti D, Nucci D, Veronese N, Fontana L, Maggi S, Onder G, Silano M, Zanetti M, Gianfredi V, Ricci M, Volpe M, Maggi S, Volpe M, Zanetti M, Onder G, Veronese N, Gianfredi V, Nucci D, Silano M, Rossi L, Antonella B, Paolo C, Maria CA, Matteo CM, Giovambattista D, Ferruccio G, Emilia G, Antonella L, Anna POS, Francesco L, Francesco M, Ida MM, Gerardo M, Barbara P, Domenico R, Elena S, Massimiliano S, Immacolata SM, Ersilia T, Andrea U, Roberto V, Giovanni Z, Elena A, Fiorella A, Letizia B, Federico B, Alberto B, Ilarj BM, Carola B, Stefania C, Ilaria C, Massimiliano C, Irene C, Amanda C, Elisa C, Anastasia C, Martina C, Giuseppe CAF, Laura C, Marco C, Stefano C, Lucilla C, Chiara D, Andrea D, Francesco DL, Lorenzo E, Maurizio F, Enrica F, Martina F, Armando F, Lucia G, Cristina G, Alessandro G, Paola G, Federica L, Fabrizia L, Elisa M, Alessandro M, Pina MM, Alessia M, Alisha M, Giovanna M, Barbara P, Stefania P, Damiano P, Cristina PM, Costanza P, Saverio RF, Cristiana R, Francesco RG, Alessia S, Piero S, Sara S, Dario TG, Ilaria T, Filippo V, Fernanda V, Marilena V, Salvatore Z, Morsella A, Limongi F, Ragusa FS, Laviano A, Sieber C (2026a)** Effectiveness of Mediterranean diet for the primary prevention of cardiovascular diseases: A systematic review and meta-analysis featured in the Italian National Guidelines “La Dieta Mediterranea”. *Nutrition*. https://doi.org/10.1016/j.nut.2025.113038
23. **Volpe R, Ciccone MM, Pala B, Barbarano F, Camastra S, Caprio M, Casirati A, Ferrera A, Galletti F, Greatti A, Mollica MP, Paolillo S, Parretti D, Nucci D, Veronese N, Fontana L, Maggi S, Onder G, Silano M, Zanetti M, Gianfredi V, Ricci M, Volpe M (2026b)** Efficacy of Mediterranean diet for the prevention of cardiovascular disease in patients: A systematic review and meta-analysis featured in the Italian National Guidelines “La Dieta Mediterranea”. *Nutrition*. https://doi.org/10.1016/j.nut.2025.113053
24. **Nordmann AJ, Suter-Zimmermann K, Bucher HC, Shai I, Tuttle KR, Estruch R, Briel M (2011)** Meta-Analysis Comparing Mediterranean to Low-Fat Diets for Modification of Cardiovascular Risk Factors. *The American Journal of Medicine*. https://doi.org/10.1016/j.amjmed.2011.04.024
25. **Trichopoulou A, Costacou T, Bamia C, Trichopoulos D (2003)** Adherence to a Mediterranean Diet and Survival in a Greek Population. *New England Journal of Medicine*. https://doi.org/10.1056/nejmoa025039
26. **Trichopoulou A, Orfanos P, Norat T, Bueno-de-Mesquita B, Ocké MC, Peeters PH, van der Schouw YT, Boeing H, Hoffmann K, Boffetta P, Nagel G, Masala G, Krogh V, Panico S, Tumino R, Vineis P, Bamia C, Naska A, Benetou V, Ferrari P, Slimani N, Pera G, Martinez-Garcia C, Navarro C, Rodriguez-Barranco M, Dorronsoro M, Spencer EA, Key TJ, Bingham S, Khaw KT, Kesse E, Clavel-Chapelon F, Boutron-Ruault MC, Berglund G, Wirfalt E, Hallmans G, Johansson I, Tjonneland A, Olsen A, Overvad K, Hundborg HH, Riboli E, Trichopoulos D (2005)** Modified Mediterranean diet and survival: EPIC-elderly prospective cohort study. *BMJ*. https://doi.org/10.1136/bmj.38415.644155.8f
27. **Sofi F, Cesari F, Abbate R, Gensini GF, Casini A (2008)** Adherence to Mediterranean diet and health status: meta-analysis. *BMJ*. https://doi.org/10.1136/bmj.a1344
28. **Eleftheriou D, Benetou V, Trichopoulou A, La Vecchia C, Bamia C (2018)** Mediterranean diet and its components in relation to all-cause mortality: meta-analysis. *British Journal of Nutrition*. https://doi.org/10.1017/s0007114518002593
29. **Soltani S, Jayedi A, Shab-Bidar S, Becerra-Tomás N, Salas-Salvadó J (2019)** Adherence to the Mediterranean Diet in Relation to All-Cause Mortality: A Systematic Review and Dose-Response Meta-Analysis of Prospective Cohort Studies. *Advances in Nutrition*. https://doi.org/10.1093/advances/nmz041
30. **Shan Z, Wang F, Li Y, Baden MY, Bhupathiraju SN, Wang DD, Sun Q, Rexrode KM, Rimm EB, Qi L, Tabung FK, Giovannucci EL, Willett WC, Manson JE, Qi Q, Hu FB (2023)** Healthy Eating Patterns and Risk of Total and Cause-Specific Mortality. *JAMA Internal Medicine*. https://doi.org/10.1001/jamainternmed.2022.6117
31. **Shan Z, Li Y, Baden MY, Bhupathiraju SN, Wang DD, Sun Q, Rexrode KM, Rimm EB, Qi L, Willett WC, Manson JE, Qi Q, Hu FB (2020)** Association Between Healthy Eating Patterns and Risk of Cardiovascular Disease. *JAMA Internal Medicine*. https://doi.org/10.1001/jamainternmed.2020.2176
32. **Ahmad S, Moorthy MV, Lee IM, Ridker PM, Manson JE, Buring JE, Demler OV, Mora S (2024)** Mediterranean Diet Adherence and Risk of All-Cause Mortality in Women. *JAMA Network Open*. https://doi.org/10.1001/jamanetworkopen.2024.14322
33. **Bonaccio M, Di Castelnuovo A, Costanzo S, Persichillo M, De Curtis A, Donati MB, de Gaetano G, Iacoviello L, on behalf of the MOLI-SANI study Investigators (2016)** Adherence to the traditional Mediterranean diet and mortality in subjects with diabetes. Prospective results from the MOLI-SANI study. *European Journal of Preventive Cardiology*. https://doi.org/10.1177/2047487315569409
34. **Hidalgo-Liberona N, Meroño T, Zamora-Ros R, Rabassa M, Semba R, Tanaka T, Bandinelli S, Ferrucci L, Andres-Lacueva C, Cherubini A (2021)** Adherence to the Mediterranean diet assessed by a novel dietary biomarker score and mortality in older adults: the InCHIANTI cohort study. *BMC Medicine*. https://doi.org/10.1186/s12916-021-02154-7
35. **Salas-Salvadó J, Bulló M, Estruch R, Ros E, Covas MI, Ibarrola-Jurado N, Corella D, Arós F, Gómez-Gracia E, Ruiz-Gutiérrez V, Romaguera D, Lapetra J, Lamuela-Raventós RM, Serra-Majem L, Pintó X, Basora J, Muñoz MA, Sorlí JV, Martínez-González MA (2014)** Prevention of Diabetes With Mediterranean Diets. *Annals of Internal Medicine*. https://doi.org/10.7326/m13-1725
36. **Martínez-González MA, Montero P, Ruiz-Canela M, Toledo E, Estruch R, Gómez-Gracia E, Li J, Ros E, Arós F, Hernáez A, Corella D, Fiol M, Lapetra J, Serra-Majem L, Pintó X, Cofán M, Sorlí JV, Babio N, Márquez-Sandoval YF, Castañer O, Salas-Salvadó J (2023)** Yearly attained adherence to Mediterranean diet and incidence of diabetes in a large randomized trial. *Cardiovascular Diabetology*. https://doi.org/10.1186/s12933-023-01994-2
37. **Konieczna J, Ruiz-Canela M, Galmes-Panades AM, Abete I, Babio N, Fiol M, Martín-Sánchez V, Estruch R, Vidal J, Buil-Cosiales P, García-Gavilán JF, Moñino M, Marcos-Delgado A, Casas R, Olbeyra R, Fitó M, Hu FB, Martínez-Gonzalez MÁ, Martínez JA, Romaguera D, Salas-Salvadó J (2023)** An Energy-Reduced Mediterranean Diet, Physical Activity, and Body Composition. *JAMA Network Open*. https://doi.org/10.1001/jamanetworkopen.2023.37994
38. **Sobiecki JG, Imamura F, Davis CR, Sharp SJ, Koulman A, Hodgson JM, Guevara M, Schulze MB, Zheng JS, Agnoli C, Bonet C, Colorado-Yohar SM, Fagherazzi G, Franks PW, Gundersen TE, Jannasch F, Kaaks R, Katzke V, Molina-Montes E, Nilsson PM, Palli D, Panico S, Papier K, Rolandsson O, Sacerdote C, Tjønneland A, Tong TYN, van der Schouw YT, Danesh J, Butterworth AS, Riboli E, Murphy KJ, Wareham NJ, Forouhi NG (2023)** A nutritional biomarker score of the Mediterranean diet and incident type 2 diabetes: Integrated analysis of data from the MedLey randomised controlled trial and the EPIC-InterAct case-cohort study. *PLOS Medicine*. https://doi.org/10.1371/journal.pmed.1004221
39. **Maroto-Rodriguez J, Ortolá R, Carballo-Casla A, Iriarte-Campo V, Salinero-Fort MÁ, Rodríguez-Artalejo F, Sotos-Prieto M (2023)** Association between a mediterranean lifestyle and Type 2 diabetes incidence: a prospective UK biobank study. *Cardiovascular Diabetology*. https://doi.org/10.1186/s12933-023-01999-x
40. **Schwingshackl L, Missbach B, König J, Hoffmann G (2015)** Adherence to a Mediterranean diet and risk of diabetes: a systematic review and meta-analysis. *Public Health Nutrition*. https://doi.org/10.1017/s1368980014001542
41. **Esposito K, Maiorino MI, Bellastella G, Chiodini P, Panagiotakos D, Giugliano D (2015)** A journey into a Mediterranean diet and type 2 diabetes: a systematic review with meta-analyses. *BMJ Open*. https://doi.org/10.1136/bmjopen-2015-008222
42. **Wu MJ, Hung CH, Yong SB, Ching GS, Hsu HJ (2025)** Impact of the Mediterranean Diet on Glycemic Control, Body Mass Index, Lipid Profile, and Blood Pressure in Type 2 Diabetes: A Meta-Analysis of Randomized Controlled Trials. *Nutrients*. https://doi.org/10.3390/nu17243908
43. **Shai I, Schwarzfuchs D, Henkin Y, Shahar DR, Witkow S, Greenberg I, Golan R, Fraser D, Bolotin A, Vardi H, Tangi-Rozental O, Zuk-Ramot R, Sarusi B, Brickner D, Schwartz Z, Sheiner E, Marko R, Katorza E, Thiery J, Fiedler GM, Blüher M, Stumvoll M, Stampfer MJ (2008)** Weight Loss with a Low-Carbohydrate, Mediterranean, or Low-Fat Diet. *New England Journal of Medicine*. https://doi.org/10.1056/nejmoa0708681
44. **Ge L, Sadeghirad B, Ball GDC, da Costa BR, Hitchcock CL, Svendrovski A, Kiflen R, Quadri K, Kwon HY, Karamouzian M, Adams-Webber T, Ahmed W, Damanhoury S, Zeraatkar D, Nikolakopoulou A, Tsuyuki RT, Tian J, Yang K, Guyatt GH, Johnston BC (2020)** Comparison of dietary macronutrient patterns of 14 popular named dietary programmes for weight and cardiovascular risk factor reduction in adults: systematic review and network meta-analysis of randomised trials. *BMJ*. https://doi.org/10.1136/bmj.m696
45. **Kastorini CM, Milionis HJ, Esposito K, Giugliano D, Goudevenos JA, Panagiotakos DB (2011)** The Effect of Mediterranean Diet on Metabolic Syndrome and its Components. *Journal of the American College of Cardiology*. https://doi.org/10.1016/j.jacc.2010.09.073
46. **Papadaki A, Nolen-Doerr E, Mantzoros CS (2020)** The Effect of the Mediterranean Diet on Metabolic Health: A Systematic Review and Meta-Analysis of Controlled Trials in Adults. *Nutrients*. https://doi.org/10.3390/nu12113342
47. **Zelicha H, Kloting N, Kaplan A, Yaskolka Meir A, Rinott E, Tsaban G, Chassidim Y, Bluher M, Ceglarek U, Isermann B, Stumvoll M, Quayson RN, von Bergen M, Engelmann B, Rolle-Kampczyk UE, Haange SB, Tuohy KM, Diotallevi C, Shelef I, Hu FB, Stampfer MJ, Shai I (2022)** The effect of high-polyphenol Mediterranean diet on visceral adiposity: the DIRECT PLUS randomized controlled trial. *BMC Medicine*. https://doi.org/10.1186/s12916-022-02525-8
48. **Muscogiuri G, Maiorino MI, Paolini B, Aversano F, Buscemi C, Cappiello I, Caruso I, Ceriani E, Chimienti M, Cicero FAG, Cintoni M, Desideri G, D'Eusebio C, Medea G, Patti MC, Randazzo C, Troiano E, Vitale M, Zimmitti S, Veronese N, Gianfredi V, Volpe M, Ricci M, Maggi S, Onder G, Silano M, Nucci D, Zanetti M, Fontana L, Casirati A, Leonardi F (2026)** Mediterranean diet for the management of pre-existing metabolic diseases: Evidence from a systematic review and meta-analysis featured in the Italian national guidelines “La Dieta Mediterranea”. *Nutrition*. https://doi.org/10.1016/j.nut.2025.112975
49. **Dinu M, Pagliai G, Casini A, Sofi F (2018)** Mediterranean diet and multiple health outcomes: an umbrella review of meta-analyses of observational studies and randomised trials. *European Journal of Clinical Nutrition*. https://doi.org/10.1038/ejcn.2017.58
50. **Morze J, Danielewicz A, Przybyłowicz K, Zeng H, Hoffmann G, Schwingshackl L (2021)** An updated systematic review and meta-analysis on adherence to mediterranean diet and risk of cancer. *European Journal of Nutrition*. https://doi.org/10.1007/s00394-020-02346-6
51. **Schwingshackl L, Schwedhelm C, Galbete C, Hoffmann G (2017)** Adherence to Mediterranean Diet and Risk of Cancer: An Updated Systematic Review and Meta-Analysis. *Nutrients*. https://doi.org/10.3390/nu9101063
52. **Zhong Y, Zhu Y, Li Q, Wang F, Ge X, Zhou G, Miao L (2020)** Association between Mediterranean diet adherence and colorectal cancer: a dose-response meta-analysis. *The American Journal of Clinical Nutrition*. https://doi.org/10.1093/ajcn/nqaa083
53. **van den Brandt PA, Schulpen M (2017)** Mediterranean diet adherence and risk of postmenopausal breast cancer: results of a cohort study and meta-analysis. *International Journal of Cancer*. https://doi.org/10.1002/ijc.30654
54. **Toledo E, Salas-Salvadó J, Donat-Vargas C, Buil-Cosiales P, Estruch R, Ros E, Corella D, Fitó M, Hu FB, Arós F, Gómez-Gracia E, Romaguera D, Ortega-Calvo M, Serra-Majem L, Pintó X, Schröder H, Basora J, Sorlí JV, Bulló M, Serra-Mir M, Martínez-González MA (2015)** Mediterranean Diet and Invasive Breast Cancer Risk Among Women at High Cardiovascular Risk in the PREDIMED Trial. *JAMA Internal Medicine*. https://doi.org/10.1001/jamainternmed.2015.4838
55. **González-Palacios Torres C, Barrios-Rodríguez R, Muñoz-Bravo C, Toledo E, Dierssen T, Jiménez-Moleón JJ (2023)** Mediterranean diet and risk of breast cancer: An umbrella review. *Clinical Nutrition*. https://doi.org/10.1016/j.clnu.2023.02.012
56. **Martínez-González MA, Sayón-Orea C, Bullón-Vela V, Bes-Rastrollo M, Rodríguez-Artalejo F, Yusta-Boyo MJ, García-Solano M (2022)** Effect of olive oil consumption on cardiovascular disease, cancer, type 2 diabetes, and all-cause mortality: A systematic review and meta-analysis. *Clinical Nutrition*. https://doi.org/10.1016/j.clnu.2022.10.001
57. **Fekete M, Varga P, Ungvari Z, Fekete JT, Buda A, Szappanos Á, Lehoczki A, Mózes N, Grosso G, Godos J, Menyhart O, Munkácsy G, Tarantini S, Yabluchanskiy A, Ungvari A, Győrffy B (2025)** The role of the Mediterranean diet in reducing the risk of cognitive impairment, dementia, and Alzheimer's disease: a meta-analysis. *GeroScience*. https://doi.org/10.1007/s11357-024-01488-3

58. **Nucci D, Sommariva A, Degoni LM, Gallo G, Mancarella M, Natarelli F, Savoia A, Catalini A, Ferranti R, Pregliasco FE, Castaldi S, Gianfredi V (2024)** Association between Mediterranean diet and dementia and Alzheimer disease: a systematic review with meta-analysis. *Aging Clinical and Experimental Research*. https://doi.org/10.1007/s40520-024-02718-6
59. **Shannon OM, Ranson JM, Gregory S, Macpherson H, Milte C, Lentjes M, Mulligan A, McEvoy C, Griffiths A, Matu J, Hill TR, Adamson A, Siervo M, Minihane AM, Muniz-Tererra G, Ritchie C, Mathers JC, Llewellyn DJ, Stevenson E (2023)** Mediterranean diet adherence is associated with lower dementia risk, independent of genetic predisposition: findings from the UK Biobank prospective cohort study. *BMC Medicine*. https://doi.org/10.1186/s12916-023-02772-3
60. **Fu J, Tan LJ, Lee JE, Shin S (2022)** Association between the mediterranean diet and cognitive health among healthy adults: A systematic review and meta-analysis. *Frontiers in Nutrition*. https://doi.org/10.3389/fnut.2022.946361
61. **Singh B, Parsaik AK, Mielke MM, Erwin PJ, Knopman DS, Petersen RC, Roberts RO (2014)** Association of Mediterranean Diet with Mild Cognitive Impairment and Alzheimer's Disease: A Systematic Review and Meta-Analysis. *Journal of Alzheimer's Disease*. https://doi.org/10.3233/jad-130830
62. **Martínez-Lapiscina EH, Clavero P, Toledo E, Estruch R, Salas-Salvadó J, San Julián B, Sanchez-Tainta A, Ros E, Valls-Pedret C, Martinez-Gonzalez MÁ (2013)** Mediterranean diet improves cognition: the PREDIMED-NAVARRA randomised trial. *Journal of Neurology, Neurosurgery & Psychiatry*. https://doi.org/10.1136/jnnp-2012-304792
63. **Barnes LL, Dhana K, Liu X, Carey VJ, Ventrelle J, Johnson K, Hollings CS, Bishop L, Laranjo N, Stubbs BJ, Reilly X, Agarwal P, Zhang S, Grodstein F, Tangney CC, Holland TM, Aggarwal NT, Arfanakis K, Morris MC, Sacks FM (2023)** Trial of the MIND Diet for Prevention of Cognitive Decline in Older Persons. *New England Journal of Medicine*. https://doi.org/10.1056/nejmoa2302368
64. **Chen H, Dhana K, Huang Y, Huang L, Tao Y, Liu X, Melo van Lent D, Zheng Y, Ascherio A, Willett W, Yuan C (2023)** Association of the Mediterranean Dietary Approaches to Stop Hypertension Intervention for Neurodegenerative Delay (MIND) Diet With the Risk of Dementia. *JAMA Psychiatry*. https://doi.org/10.1001/jamapsychiatry.2023.0800
65. **Gregory S, Buller-Peralta I, Bridgeman K, Góngora VDLC, Dounavi M, Low A, Ntailianis G, O'Brien J, Parra MA, Ritchie CW, Ritchie K, Shannon OM, Stevenson EJ, Muniz-Terrera G (2024)** The Mediterranean diet is not associated with neuroimaging or cognition in middle-aged adults: a cross-sectional analysis of the PREVENT dementia programme. *European Journal of Neurology*. https://doi.org/10.1111/ene.16345
66. **Lassale C, Batty GD, Baghdadli A, Jacka F, Sánchez-Villegas A, Kivimäki M, Akbaraly T (2019)** Healthy dietary indices and risk of depressive outcomes: a systematic review and meta-analysis of observational studies. *Molecular Psychiatry*. https://doi.org/10.1038/s41380-018-0237-8
67. **Shafiei F, Salari-Moghaddam A, Larijani B, Esmaillzadeh A (2019)** Adherence to the Mediterranean diet and risk of depression: a systematic review and updated meta-analysis of observational studies. *Nutrition Reviews*. https://doi.org/10.1093/nutrit/nuy070
68. **Jacka FN, O'Neil A, Opie R, Itsiopoulos C, Cotton S, Mohebbi M, Castle D, Dash S, Mihalopoulos C, Chatterton ML, Brazionis L, Dean OM, Hodge AM, Berk M (2017)** A randomised controlled trial of dietary improvement for adults with major depression (the 'SMILES' trial). *BMC Medicine*. https://doi.org/10.1186/s12916-017-0791-y
69. **Parletta N, Zarnowiecki D, Cho J, Wilson A, Bogomolova S, Villani A, Itsiopoulos C, Niyonsenga T, Blunden S, Meyer B, Segal L, Baune BT, O'Dea K (2019)** A Mediterranean-style dietary intervention supplemented with fish oil improves diet quality and mental health in people with depression: A randomized controlled trial (HELFIMED). *Nutritional Neuroscience*. https://doi.org/10.1080/1028415x.2017.1411320
70. **Bayes J, Schloss J, Sibbritt D (2022)** The effect of a Mediterranean diet on the symptoms of depression in young males (the "AMMEND: A Mediterranean Diet in MEN with Depression" study): a randomized controlled trial. *The American Journal of Clinical Nutrition*. https://doi.org/10.1093/ajcn/nqac106
71. **Bizzozero-Peroni B, Martínez-Vizcaíno V, Fernández-Rodríguez R, Jiménez-López E, Núñez de Arenas-Arroyo S, Saz-Lara A, Díaz-Goñi V, Mesas AE (2025)** The impact of the Mediterranean diet on alleviating depressive symptoms in adults: a systematic review and meta-analysis of randomized controlled trials. *Nutrition Reviews*. https://doi.org/10.1093/nutrit/nuad176
72. **Kabthymer RH, Karimi L, Livesay K, Lee M, Apostolopoulos V, Millar R, McKay S, Barry S, Olga CN, Colomer MF, Soultanakis H, Conduit R, Takac M, Mizzi S, Sidossis LS, Tierney A, Itsiopoulos C, Feehan J, Courten BD (2026)** Effect of Mediterranean diet on mental health outcomes: a systematic review. *Nutrition Research Reviews*. https://doi.org/10.1017/s0954422425100243
73. **Properzi C, O'Sullivan TA, Sherriff JL, Ching HL, Jeffrey GP, Buckley RF, Tibballs J, MacQuillan GC, Garas G, Adams LA (2018)** Ad Libitum Mediterranean and Low-Fat Diets Both Significantly Reduce Hepatic Steatosis: A Randomized Controlled Trial. *Hepatology*. https://doi.org/10.1002/hep.30076
74. **Gepner Y, Shelef I, Komy O, Cohen N, Schwarzfuchs D, Bril N, Rein M, Serfaty D, Kenigsbuch S, Zelicha H, Yaskolka Meir A, Tene L, Bilitzky A, Tsaban G, Chassidim Y, Sarusy B, Ceglarek U, Thiery J, Stumvoll M, Blüher M, Stampfer MJ, Rudich A, Shai I (2019)** The beneficial effects of Mediterranean diet over low-fat diet may be mediated by decreasing hepatic fat content. *Journal of Hepatology*. https://doi.org/10.1016/j.jhep.2019.04.013
75. **Yaskolka Meir A, Rinott E, Tsaban G, Zelicha H, Kaplan A, Rosen P, Shelef I, Youngster I, Shalev A, Blüher M, Ceglarek U, Stumvoll M, Tuohy K, Diotallevi C, Vrhovsek U, Hu F, Stampfer M, Shai I (2021)** Effect of green-Mediterranean diet on intrahepatic fat: the DIRECT PLUS randomised controlled trial. *Gut*. https://doi.org/10.1136/gutjnl-2020-323106
76. **Kawaguchi T, Charlton M, Kawaguchi A, Yamamura S, Nakano D, Tsutsumi T, Zafer M, Torimura T (2021)** Effects of Mediterranean Diet in Patients with Nonalcoholic Fatty Liver Disease: A Systematic Review, Meta-Analysis, and Meta-Regression Analysis of Randomized Controlled Trials. *Seminars in Liver Disease*. https://doi.org/10.1055/s-0041-1723751
77. **Sangouni AA, Hassani Zadeh S, Mozaffari-Khosravi H, Hosseinzadeh M (2022)** Effect of Mediterranean diet on liver enzymes: a systematic review and meta-analysis of randomised controlled trials. *British Journal of Nutrition*. https://doi.org/10.1017/s0007114521002270
78. **Xiong Y, Shi X, Xiong X, Li S, Zhao H, Song H, Wang J, Zhang L, You S, Ji G, Liu B, Wu N (2024)** A systematic review and meta-analysis of randomized controlled trials: effects of mediterranean diet and low-fat diet on liver enzymes and liver fat content of NAFLD. *Food & Function*. https://doi.org/10.1039/d4fo01461h
79. **Chooi YC, Zhang QA, Magkos F, Ng M, Michael N, Wu X, Volchanskaya VSB, Lai X, Wanjaya ER, Elejalde U, Goh CC, Yap CPL, Wong LH, Lim KJ, Velan SS, Yaligar J, Muthiah MD, Chong YS, Loo EXL, Eriksson JG, Lim KLM, Kouk MSF, Mei Chong EW, Gani MA, Li L, Tay VHK, Kway YM, Kumar M, Sadananthan SA, Khoo K, Koh D, Lim R, Kang CW, Sin KL, Lim JW (2024)** Effect of an Asian-adapted Mediterranean diet and pentadecanoic acid on fatty liver disease: the TANGO randomized controlled trial. *The American Journal of Clinical Nutrition*. https://doi.org/10.1016/j.ajcnut.2023.11.013
80. **Schwingshackl L, Hoffmann G (2014)** Mediterranean dietary pattern, inflammation and endothelial function: A systematic review and meta-analysis of intervention trials. *Nutrition, Metabolism and Cardiovascular Diseases*. https://doi.org/10.1016/j.numecd.2014.03.003
81. **Koelman L, Egea Rodrigues C, Aleksandrova K (2022)** Effects of Dietary Patterns on Biomarkers of Inflammation and Immune Responses: A Systematic Review and Meta-Analysis of Randomized Controlled Trials. *Advances in Nutrition*. https://doi.org/10.1093/advances/nmab086
82. **Keshani M, Rafiee S, Heidari H, Rouhani MH, Sharma M, Bagherniya M (2026)** Mediterranean Diet Reduces Inflammation in Adults: A Systematic Review and Meta-analysis of Randomized Controlled Trials. *Nutrition Reviews*. https://doi.org/10.1093/nutrit/nuaf213
83. **Itsiopoulos C, Mayr HL, Thomas CJ (2022)** The anti-inflammatory effects of a Mediterranean diet: a review. *Current Opinion in Clinical Nutrition & Metabolic Care*. https://doi.org/10.1097/mco.0000000000000872
84. **Meslier V, Laiola M, Roager HM, De Filippis F, Roume H, Quinquis B, Giacco R, Mennella I, Ferracane R, Pons N, Pasolli E, Rivellese A, Dragsted LO, Vitaglione P, Ehrlich SD, Ercolini D (2020)** Mediterranean diet intervention in overweight and obese subjects lowers plasma cholesterol and causes changes in the gut microbiome and metabolome independently of energy intake. *Gut*. https://doi.org/10.1136/gutjnl-2019-320438
85. **Ghosh TS, Rampelli S, Jeffery IB, Santoro A, Neto M, Capri M, Giampieri E, Jennings A, Candela M, Turroni S, Zoetendal EG, Hermes GDA, Elodie C, Meunier N, Brugere CM, Pujos-Guillot E, Berendsen AM, De Groot LCPGM, Feskins EJM, Kaluza J, Pietruszka B, Bielak MJ, Comte B, Maijo-Ferre M, Nicoletti C, De Vos WM, Fairweather-Tait S, Cassidy A, Brigidi P, Franceschi C, O'Toole PW (2020)** Mediterranean diet intervention alters the gut microbiome in older people reducing frailty and improving health status: the NU-AGE 1-year dietary intervention across five European countries. *Gut*. https://doi.org/10.1136/gutjnl-2019-319654
86. **Haro C, Montes-Borrego M, Rangel-Zúñiga OA, Alcalá-Díaz JF, Gómez-Delgado F, Pérez-Martínez P, Delgado-Lista J, Quintana-Navarro GM, Tinahones FJ, Landa BB, López-Miranda J, Camargo A, Pérez-Jiménez F (2016)** Two Healthy Diets Modulate Gut Microbial Community Improving Insulin Sensitivity in a Human Obese Population. *The Journal of Clinical Endocrinology & Metabolism*. https://doi.org/10.1210/jc.2015-3351
87. **Li J, Guasch-Ferré M, Chung W, Ruiz-Canela M, Toledo E, Corella D, Bhupathiraju SN, Tobias DK, Tabung FK, Hu J, Zhao T, Turman C, Feng YCA, Clish CB, Mucci L, Eliassen AH, Costenbader KH, Karlson EW, Wolpin BM, Ascherio A, Rimm EB, Manson JE, Qi L, Martínez-González MÁ, Salas-Salvadó J, Hu FB, Liang L (2020)** The Mediterranean diet, plasma metabolome, and cardiovascular disease risk. *European Heart Journal*. https://doi.org/10.1093/eurheartj/ehaa209
88. **Gao P, Rinott E, Dong D, Mei Z, Wang F, Liu Y, Kamer O, Yaskolka Meir A, Tuohy KM, Blüher M, Stumvoll M, Stampfer MJ, Shai I, Wang DD (2024)** Gut microbial metabolism of bile acids modifies the effect of Mediterranean diet interventions on cardiometabolic risk in a randomized controlled trial. *Gut Microbes*. https://doi.org/10.1080/19490976.2024.2426610
89. **García-Gavilán JF, Atzeni A, Babio N, Liang L, Belzer C, Vioque J, Corella D, Fitó M, Vidal J, Moreno-Indias I, Torres-Collado L, Coltell O, Toledo E, Clish C, Hernando J, Yun H, Hernández-Cacho A, Jeanfavre S, Dennis C, Gómez-Pérez AM, Martínez MA, Ruiz-Canela M, Tinahones FJ, Hu FB, Salas-Salvadó J (2024)** Effect of 1-year lifestyle intervention with energy-reduced Mediterranean diet and physical activity promotion on the gut metabolome and microbiota: a randomized clinical trial. *The American Journal of Clinical Nutrition*. https://doi.org/10.1016/j.ajcnut.2024.02.021
90. **Hoffman R, Gerber M (2013)** Evaluating and adapting the Mediterranean diet for non-Mediterranean populations: A critical appraisal. *Nutrition Reviews*. https://doi.org/10.1111/nure.12040
91. **Woodside J, Young IS, McKinley MC (2022)** Culturally adapting the Mediterranean Diet pattern – a way of promoting more 'sustainable' dietary change? *British Journal of Nutrition*. https://doi.org/10.1017/s0007114522001945
92. **Jennings A, Shannon OM, Gillings R, Lee V, Elsworthy R, Bundy R, Rao G, Hanson S, Hardeman W, Paddick SM, Siervo M, Aldred S, Mathers JC, Hornberger M, Minihane AM (2024)** Effectiveness and feasibility of a theory-informed intervention to improve Mediterranean diet adherence, physical activity and cognition in older adults at risk of dementia: the MedEx-UK randomised controlled trial. *BMC Medicine*. https://doi.org/10.1186/s12916-024-03815-z
93. **Biggi C, Biasini B, Ogrinc N, Strojnik L, Endrizzi I, Menghi L, Khémiri I, Mankai A, Slama FB, Jamoussi H, Riviou K, Elfazazi K, Rehman N, Scazzina F, Menozzi D (2024)** Drivers and Barriers Influencing Adherence to the Mediterranean Diet: A Comparative Study across Five Countries. *Nutrients*. https://doi.org/10.3390/nu16152405
94. **Lorca-Camara V, Bosque-Prous M, Bes-Rastrollo M, O'Callaghan-Gordo C, Bach-Faig A (2024)** Environmental and Health Sustainability of the Mediterranean Diet: A Systematic Review. *Advances in Nutrition*. https://doi.org/10.1016/j.advnut.2024.100322
95. **Bonekamp NE, Visseren FLJ, van der Schouw YT, van der Meer MG, Teraa M, Ruigrok YM, Geleijnse JM, Koopal C, UCC-SMART study group, Cramer MJ, Nathoe HM, van der Harst P, van de Meer MG, de Borst GJ, Teraa M, Bots ML, van Smeden M, Emmelot-Vonk MH, de Jong PA, Lely AT, van der Kaaij NP, Kappelle LJ, Ruigrok YM, Verhaar MC, Dorresteijn JAN, Visseren FLJ (2024)** Cost-effectiveness of Mediterranean diet and physical activity in secondary cardiovascular disease prevention: results from the UCC-SMART cohort study. *European Journal of Preventive Cardiology*. https://doi.org/10.1093/eurjpc/zwae123
96. **Dernini S, Berry E, Serra-Majem L, La Vecchia C, Capone R, Medina F, Aranceta-Bartrina J, Belahsen R, Burlingame B, Calabrese G, Corella D, Donini L, Lairon D, Meybeck A, Pekcan A, Piscopo S, Yngve A, Trichopoulou A (2017)** Med Diet 4.0: the Mediterranean diet with four sustainable benefits. *Public Health Nutrition*. https://doi.org/10.1017/s1368980016003177
97. **Martínez-González MA (2024)** Should we remove wine from the Mediterranean diet?: a narrative review. *The American Journal of Clinical Nutrition*. https://doi.org/10.1016/j.ajcnut.2023.12.020 ■